

Perturbations in Earth's Atmosphere over An Indian Region during the Total Solar Eclipse on 22 July 2009

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ABSTRACT

During a total solar eclipse (TSE) on 22 July 2009, atmospheric perturbations were monitored from the surface to thermosphere to understand TSE's impact on the meteorological (temperature, relative humidity, wind speed, and wind direction) and chemical (O_3 and NO_x) parameters around Kadapa ($14.28^\circ N$, $78.42^\circ E$), a tropical semi-arid region of India. For this purpose, an experiment was conducted at Yogi Vemana University Campus, Kadapa, India, to measure the temperature, wind speed, wind direction, and concentrations of ozone (O_3), NO, NO_2 , and NO_x by using the automatic weather station (AWS) and O_3 analyzer. On the eclipse day (22 July 2009), the surface observations at Kadapa showed a reduction in temperature (about $1.1^\circ C$) because of the solar insulation. Comparison of the thermal, dynamical (wind), and chemical parameters on the TSE day with control days [preceding (21 July 2009) and succeeding (23 July 2009) the TSE] illustrated the influence of solar eclipse. During the eclipse period, the O_3 mixing ratio decreased, while NO_2 and NO_x increased; however, NO remained unchanged. In addition, radio occultation (RO) temperature profiles from Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC)/Formosat Satellite Mission (FORMOSAT-3) and Thermosphere, Ionosphere, and Mesosphere Energetics and Dynamics (TIMED) satellites were utilized to understand the impact of TSE on dynamics of the middle and upper atmosphere from tropopause to the thermosphere. High vertical resolution COSMIC observations revealed that during the solar eclipse, tropopause was cooler with twin peaks (double tropopause). The lower thermosphere between 110 and 130 km became warmer during the TSE, which might be caused by the dynamical response of the atmosphere in this region to the solar eclipse. The experimental data have provided very fine-scale variations of the atmospheric parameters both in time and height and also constituted a new set of results on TSE for further research.

Key words: total solar eclipse (TSE), atmospheric perturbations, tropopause dynamics, COSMIC/FORMOSAT-3 satellite, radio occultation (RO) observations

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1. Introduction

Total solar eclipse (TSE) is an unusual natural phenomenon that occurs at a particular location because totality exists only along a narrow path on the earth's surface traced by the moon's full shadow or umbra (Szałowski, 2002). TSE occurs at a particular location fast and provides a chance to study the interaction of ionizing radiations with the earth's surface due to the rapid changes in radiation (Kumar and Rengaiyan, 2011). Sev-

eral researchers investigated the effect of solar eclipses on various atmospheric/meteorological, chemical, and dynamical parameters from the lower atmosphere to ionospheric region (Cohen, 1984; Singh et al., 1989; Abraham et al., 1998; Boitman et al., 1999; Farges et al., 2001; Dutta et al., 2011). Abraham et al. (1998) presented the absorption of radio wave (2.5 MHz) in the ionospheric region at Delhi when a partial solar eclipse occurred on 24 October 1995, and pointed out that the minimum absorption of data in the ionosphere was observed

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four minutes earlier than the ground level eclipse totality. Phanikumar et al. (2014) have shown a linear relationship of the electron density of D-region and solar radiation using very low frequency (VLF) transmitter signal amplitude data. Meanwhile, Maurya et al. (2014) discussed the gravity waves (GWs) structure in the ionospheric region during the solar eclipse period. They suggested that the solar eclipse induced GWs propagating into ionosphere had a source both from bottom and top. They investigated that during the eclipse, GWs generated at the altitude of about 200 km between F1 and F2 regions of the ionosphere propagated downward to the D-region. The downward propagation of solar eclipse induced GWs may have an impact on the thermal structure of lower region, which needs to be investigated in a more sophisticated way. The present study will investigate the change in the temperature structure from surface to the thermosphere during the TSE on 22 July 2009.

During a solar eclipse, the variation of different surface meteorological parameters in tropospheric and stratospheric regions has been a most challenging problem (Muraleedharan et al., 2011). Some previous observational studies on solar eclipse have examined the changes of meteorological parameters, such as wind, air temperature, atmospheric pressure, humidity, GWs, and ozone (O_3) measurements (Chimonas and Hines, 1971; Anderson et al., 1972; Chakrabarty et al., 1997; Zerefos et al., 2000; Ramchandran et al., 2002; Szałowski, 2002; Krishnan et al., 2004; Tzanis et al., 2008; Nymphas et al., 2009). These studies concentrated either on the lower atmosphere or middle/upper atmosphere. However, so far the studies on the impacts induced by solar eclipses on the lower and middle atmosphere are still quite rudimentary. By using fine-scale data from *Thermosphere, Ionosphere, and Mesosphere Energetics and Dynamics (TIMED)* and Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC)/*Formosat Satellite Mission (FORMOSAT-3)*, the current study tries to present new dynamics of the lower and middle atmosphere during the solar eclipse on 22 July 2009.

This solar eclipse event also provides an excellent opportunity to study the temporal response of earth's surface to an abrupt change in solar radiation. During different solar eclipses over India, various measurements were made at different locations, as documented in studies by Appu et al. (1997), Dutta et al. (1999), and Krishnan et al. (2004). These studies reported the changes in various surface meteorological parameters. Some of the investigations concentrated on the boundary layer, and temperature and O_3 measurements in the lower stratosphere (Narasimha et al., 1982; Ratnam et al., 2010, 2011; Namboodiri et al., 2011; Subrahmanyam and Anurose, 2011;

Bhat and Jagannathan, 2012). These previous studies indicated that there exist several different types of thermal and chemical perturbations in the earth's atmosphere, but their vertical distribution is rarely understood in the stratospheric and mesospheric regions. Subrahmanyam et al. (2011) investigated the solar eclipse induced perturbation of temperature in the upper tropospheric and lower stratospheric (UTLS) region. They revealed that during the partial solar eclipse, the temperature decreased by 2–3°C in the vicinity of tropopause while it increased by approximately 2.6°C in the lower stratosphere during the maximum phase. Their study did not provide any information on the temperature variability at the surface. Later, Rao et al. (2013) investigated the impact of TSE on the surface layer processes and tropospheric temperature structure at Dibrugarh, the northeast region of India, which came under the path of totality of the eclipse (Fig. 1). They also showed a significant reduction in the wind speed during the eclipse window, a clear reversal of zonal wind direction, and its sign reversal from westerly to easterly.

Some of previous studies concentrated on the variation of chemical parameters (NO_2 and NO) during different solar eclipses at different locations (Nishanth et al. 2011). Tzanis et al. (2008) pointed out that the NO_2 response to solar eclipse would depend upon stations, and the time lag of NO_2 response to eclipse was different for different stations. This study mainly focused on the time lag response of NO_2 . Another study by Sharma et al. (2010) during the solar eclipse of 15 January 2010 reported a decrease in NO_2 , while on the contrary a study by Nishanth et al. (2011) for the same solar eclipse showed an increase in the NO_2 concentration. To resolve such discrepancy, more observational studies are required. In the current study, we will try to address the reasons for such contradictions in the previous studies.

In spite of the intensive research carried out on solar eclipses during the past few decades, the solar eclipse induced thermal and chemical variability still requires fine-scale observational study. Due to aforesaid reasons, for the first time in India, multi-sensor data are utilized in this investigation to understand the chemical and thermal response of TSE of 22 July 2009 from the surface to thermosphere at the fine-scale level.

Moreover, most of the earlier solar eclipses that have been studied occurred mostly in the mid afternoon and late evening, while the TSE on 22 July 2009 occurred during the early morning hours in the Indian region and lasted several minutes (4 min 39 s) all along the path over northern India, northern Bangladesh, and eastern Nepal. After exiting the mainland Asia, the TSE's path crossed Japan's Ryukyu Islands and curved southeast-

ward through the Pacific Ocean, where the maximum duration of totality reached 6 min 39 s. This solar eclipse, as a part of Solar Saros series 136, is the longest one of the century till now; in fact, the totality is of the longest duration in any solar eclipse between 1991 and 2132. Details related to the time periods of all solar eclipses up to 2200 can be found at the NASA website <https://eclipse.gsfc.nasa.gov/SEcat5/SE2101-2200.html>. The previous solar eclipse of Saros 136 occurred on 11 July 1991, which was slightly longer than the current one, lasting 6 min 53 s. Other previous eclipses of the same series occurred on 30 June 1973 and 20 June 1955, which were longer, lasting 7 min 4 s and 7 min 8 s, respectively. The future eclipse event from this series will be on 2 August 2027, but the next one that will surpass the current one in duration is due on 13 June 2132 only.

Most of the above mentioned studies focused on the location, which came under the path of totality of solar eclipse. Note that the eclipse on 22 July 2009 over passed at northern India though our observational site at Kadapa nearly 1500 km away from the main path. From the fluid dynamics point of view, changes taking place all of a sudden at a particular altitude level cause responses in other places due to the fluid characteristics of the atmosphere. Such a rapid connection from one location to another, even from polar to tropical regions, in the atmosphere, was identified in the case of sudden stratospheric warming event (Dhaka et al., 2015). Thus, it is quite rational to study the influence of solar eclipse on the atmospheric dynamics at the stations located far from the path of the solar eclipse.

2. Experimental location and data

An Automatic Weather Station (AWS) for measurement of temperature, wind, and humidity, also with O_3 , NO , NO_2 , and NO_x sensors, has been set up over Kadapa ($14.28^\circ N$, $78.42^\circ E$), India. Note that the total number of AWSs in India is 1611. A complete distribution of AWSs over the whole India can be seen at <http://www.mosdac.gov.in/data/insituBrowse.do?mode=AwsDisributionMap>. The AWS located at Kadapa is far better and advantageous due to its high temporal resolution of data. The AWS at Kadapa gives 1-min average data, whereas others provide data on the hourly basis. In order to examine the solar eclipse induced effects on the surface level at Kadapa, we utilize the AWS data of surface temperature, relative humidity, and wind speed around the eclipse period on 22 July 2009 (i.e., 20–24 July 2009). Along with these surface observations, we also use the upper air observations, i.e., the radiosonde data, from the Indian Meteorological Department, at Hyderabad ($17.4^\circ N$,

$78.2^\circ E$), which is spatially about 300 km from Kadapa, at 0530 Indian Standard Time (IST). The radiosonde provides vertical profiles of different meteorological parameters (temperature, pressure, humidity, etc.) during the observational period (ascent time). The temperature accuracy is less than 1 K. The time period of solar eclipse (22 July 2009) fortunately coincided approximately with the launching time of the radiosonde.

To understand the impact of solar eclipse over the entire Indian region (65° – $110^\circ E$), we investigated the temperature profiles from Global Positioning System (GPS) Radio Occultation (RO) satellite data from COSMIC/*FORMOSAT-3* (hereafter COSMIC). The COSMIC mission is a collaborative project of the “National” Space Organization in Taiwan, China and the University Corporation for Atmospheric Research (UCAR) in the USA. The COSMIC mission is a first RO mission to use a constellation of six satellites in the Low Earth Orbit (LEO). The COSMIC RO has provided accurate measurements of temperature from 5- to 40-km altitudes (Ho et al., 2009, 2010a, b). It has a wide spatial coverage with high vertical resolution (from about 300 m near the surface to about 1.5 km at 40 km) and accuracy (equivalent to < 1 K; average accuracy < 0.1 K; Ho et al., 2009, 2010a, b; Kumar et al., 2014, 2017). These data profiles are interpolated at the 100-m height interval. For this study, the level-2 wet atmospheric profiles are obtained from COSMIC/*FORMOSAT-3* (available at www.cosmic.ucar.edu).

In addition to AWS, radiosonde, and COSMIC data, the data from Sounding of the Atmosphere using Broad Emission Radiometry (SABER) instrument onboard the *TIMED* satellite are also used to study the solar eclipse induced perturbations in the thermosphere. SABER is a limb viewing infrared radiometer that measures the thermal structure and composition of the atmosphere from the upper troposphere (20 km) into lower thermosphere (120 km) and one of the four instruments on board the *TIMED* satellite launched by NASA (Mlynarczyk, 1997; Russell et al., 1999). The SABER data have good temperature accuracy, with errors of the order of ± 1.4 K in the lower stratosphere, ± 1 K in the middle stratosphere, and ± 2 K in the upper stratosphere and lower thermosphere (Remsberg et al., 2008).

We confined our study only to the Indian longitude between 65° and $110^\circ E$ rather than the entire eclipse path. To investigate the impact of TSE, we searched the GPS RO data for every 1-min eclipse location that occurred within a 3-h time window (1.5 h before and after the passage of eclipse; Espenak and Anderson, 2008). Time 1.5 h almost corresponds to the 2250-km distance

in space (3-h time difference approximately corresponds to the 45-degree difference in longitude). The path of the TSE on 22 July 2009 over the Indian region along with satellite data from COSMIC and SABER is shown in Fig. 1. Blue and red lines represent the edges and center of moon's umbra shadow during the eclipse period, respectively. The blue filled circles in Fig. 1a indicate GPS RO profiles that are located within a 3-h time window along the 1-min resolution eclipse path and also within a 2250-km radius from the center of eclipse location. Figure 1b represents the soundings from SABER data. These daily GPS RO profiles were averaged for 22 July 2009. In order to compare the daily GPS RO profiles during the solar eclipse to total (control) days (20–24 July), we used the same eclipse path, time window, and spatial radius during the total (control) days. From 20 to 24 July 2009, the total number of GPS-RO occultations and their timing of occurrence along with the longitude and latitude position are shown in Table 1.

3. Results and discussion

3.1 Surface meteorological parameters

Each solar eclipse, depending upon its time of occur-

rence, duration, and geographical location, has a unique effect on the earth's atmosphere. TSE provides a rare opportunity to study the response of earth's atmosphere to a step function kind of input. Obscuration (magnitude) of solar eclipse is the fraction of sun's area (diameter) occulted by the moon. The TSE of 22 July 2009 occurred with the obscuration of 69%. This solar eclipse was initiated at 0530 IST over Kadapa and the maximum phase occurred at 0621 IST. It ended over Kadapa at 0718 IST. During the solar eclipse, the incoming solar radiation was obscured by the moon, and significant changes occurred in meteorological parameters from the surface to thermosphere. Hence, we investigate the effect of solar eclipse on various surface meteorological parameters over Kadapa during the 22 July 2009 solar eclipse, and on the proceeding and succeeding days/control days (i.e., 20, 21, 23, and 24 July). The mean variation of temperature, relative humidity, wind speed, and wind direction along with standard deviation are shown in Fig. 2. Figure 2a illustrates the temporal variation of air temperature, while Figs. 2b–d represent the temporal variations of relative humidity, wind speed, and wind direction, respectively. Three vertical lines marked at 0530, 0645, and 0800 IST in Figs. 2b–d indicate the timings of beginning, peak, and ending of TSE, respectively. It is evident from Fig. 2 that

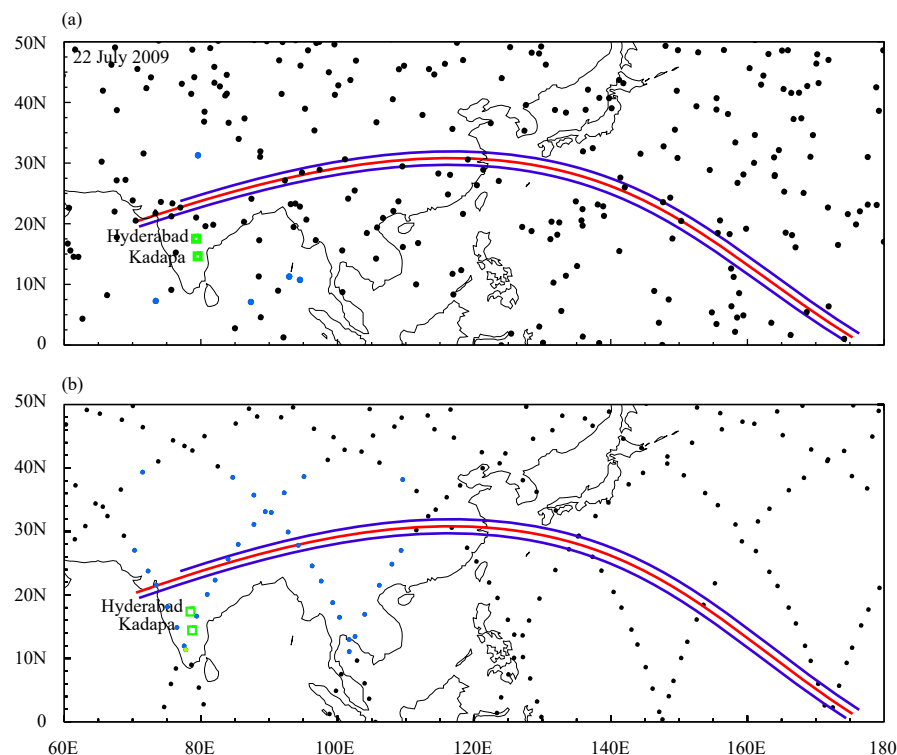
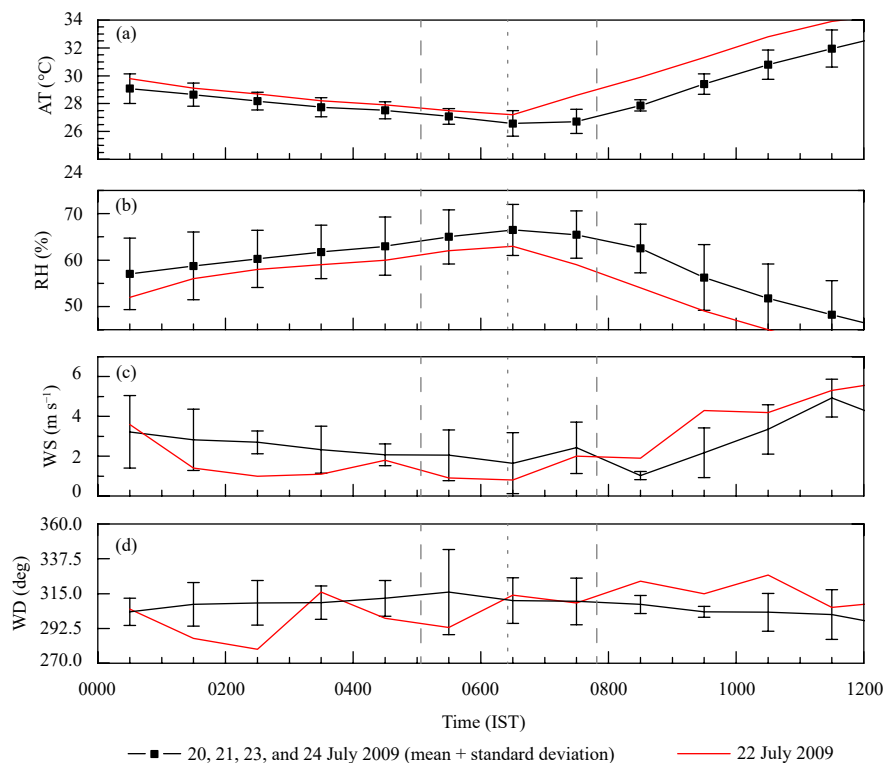


Fig. 1. The solar eclipse (22 July 2009) path over the Indian and Indonesian regions. The blue and red lines represent the edges and center of moon's umbra shadow respectively during the eclipse period. The dots in (a) and (b) represent occultation points of the COSMIC satellite and *TIMED*/SABER orbits respectively. The blue dots in (a) and (b) are located within a 3-h time window along the 1-min resolution eclipse path and also within a 2250-km radius from the center of eclipse location.

Table 1. The total number of GPS-RO occultations and their timing of occurrence along with the longitude and latitude position during 20–24 July 2009

Day	Number of occultations	Time (UTC)	Latitude	Longitude
20 July 2009	8	0015; 0203	20.01°N; 09.49°N	95.73°E; 95.70°E
		0119; 0228	03.68°N; 41.70°N	78.94°E; 79.05°E
		0012; 0016	14.64°N; 20.75°N	70.66°E; 102.61°E
		0143; 0150	15.15°N; 24.39°N	74.35°E; 73.23°E
21 July 2009	10	0003; 0127	08.48°N; 32.48°N	88.30°E; 79.62°E
		0135; 0141	09.66°N; 21.93°N	95.40°E; 100.52°E
		0305; 0013	11.34°N; 11.47°N	78.15°E; 70.64°E
		0023; 0059	15.10°N; 47.50°N	84.57°E; 101.99°E
		0252; 0130	10.79°N; 20.12°N	98.78°E; 69.98°E
22 July 2009	5	0106; 0107	11.28°N; 10.74°N	92.93°E; 94.50°E
		0111; 0053	07.26°N; 31.27°N	73.4°E; 79.57°E;
		0057	07.10°N	87.32°E
23 July 2009	10	0036; 0038	22.47°N; 12.99°N	93.03°E; 93.05°E
		0047; 0051	22.66°N; 12.03°N	72.93°E; 109.73°E
		0229; 0015	21.07°N; 18.80°N	82.73°E; 79.13°E
		0004; 0324	45.80°N; 31.84°N	103.17°E; 104.95°E
24 July 2009	8	0314; 0314	06.94°N; 44.96°N	84.23°E; 95.92°E
		0006; 0010	33.56°N; 16.60°N	92.66°E; 91.05°E
		0300; 0253	45.57°N; 42.93°N	107.91°E; 103.42°E
		0311; 0316	12.03°N; 08.61°N	83.38°E; 96.51°E
		0008; 0017	17.57°N; 46.35°N	94.06°E; 94.10°E

Note: IST = UTC + 0530.

**Fig. 2.** Temporal variations of (a) air temperature (AT; °C), (b) relative humidity (RH; %), (c) wind speed (WS; m s^{-1}), and (d) wind direction (WD; degree). The red and black lines represent the solar eclipse day (22 July) and mean of control days (20, 21, 23, and 24 July 2009) with the standard deviation, respectively. The three vertical dashed lines marked at 0530, 0645, and 0800 IST indicate the timings of beginning, peak, and ending of the total solar eclipse (TSE), respectively.

significant changes in the surface temperature, relative humidity, wind speed, and wind direction can be seen during the solar eclipse compared to the control days. The surface temperature during the eclipse followed nor-

mal diurnal behaviors until 0700 IST, but after that a higher deviation occurred in comparison to control days. This increase in the air temperature is nearly 1°C (Fig. 2a). Most of previous studies in India, Greece, and South

Korea have shown that the surface temperature decreased about 0.5 to 1.2°C during the solar eclipse (Ramchandran et al., 2002; Krishnan et al., 2004; Founda et al., 2007; Bhattacharya et al., 2010; Chung et al., 2010). But interestingly, we observed an increase in air temperature by about 1°C during and after the eclipse. Similarly, we obtained the contrast perturbation (decrease) in the relative humidity pattern in comparison to previous studies. Muraleedharan et al. (2011) reported that the relative humidity increased during the eclipse but we have observed a decrease by around 3% (Fig. 2b). It is noteworthy from Fig. 2b that the temporal variation of humidity shows a variation in contrast during and after the solar eclipse. The wind speed also shows a response to solar eclipse (Fig. 2c). During the solar eclipse days, the mean wind speed is 1.5 m s⁻¹ between 0530 and 0800 IST while it is 2.5 m s⁻¹ on total control days for the same time period (Fig. 2c). Hence, a decrease of approximately 1.0 m s⁻¹ in the mean wind speed was observed during the solar eclipse. However, after the solar eclipse time, the temporal variation of wind speed shows a higher value in comparison to total control days. The wind direction is shown in Fig. 2d. The physical interpretation of the degree in terms of the direction can be understood as: 270° denotes west, 315° northwest, 337.5° north-northwest, and 360° north. From Fig. 2d, we can clearly see that there are perturbations in the wind direction on the eclipse day in comparison to control days, and continuous periodical changes are emerging in the wind direction. These changes in the wind speed and directions may play an important role in affecting the concentration of NO_x (NO₂ + NO) during the eclipse, which will be discussed in the subsequent section.

As discussed above, we obtained contrast variation patterns in the surface temperature and relative humidity

in comparison to previous studies. Note that previous studies mainly focused on the location that came under the path of totality of solar eclipse. Our location Kadapa, a semi-arid region, does not come under the totality of solar eclipse on 22 July 2009 but slightly away from the eclipse path. This study points out that each geographical location has a unique impact of eclipse. In the next section, we will discuss the impact of solar eclipse on chemical parameters.

3.2 Influence of solar eclipse on chemical parameters (O₃, NO, NO₂, and NO_x)

3.2.1 Impact on O₃

The mixing ratio of surface O₃ was measured by using the O₃ analyzer for the period of 21–23 July 2009 over Kadapa. Figure 3 represents the temporal variation of surface O₃ during 0000–2400 IST from 21 to 23 July 2009. The day time increase in O₃ between 0800 and 1600 IST is quite visible during all the days. Daytime increase of the O₃ mixing ratio is basically due to the photo-oxidation of precursor gases, like CO, CH₄, and other hydrocarbons in the presence of sufficient amount of NO/NO_x. It is noticeable in Fig. 3 that on the pre-solar eclipse day (21 July 2009), the average mixing ratio of surface O₃ is recorded as 36.02 ± 6.68 ppbv within a range of 28.2–48.0 ppbv; whereas on the eclipse day, the average mixing ratio of surface O₃ is recorded as 34.80 ± 7.39 ppbv within a range of 20.0–46.0 ppbv. Again on the succeeding day (23 July 2009), the variations in surface O₃ resembled with those on 21 July 2009. On the TSE day, i.e., 22 July 2009, the mixing ratio of O₃ decreased slightly with the beginning of solar eclipse and reached a minimum of 20 ppbv at 0730 IST (nearly the peak period of eclipse). After the TSE time, the mixing

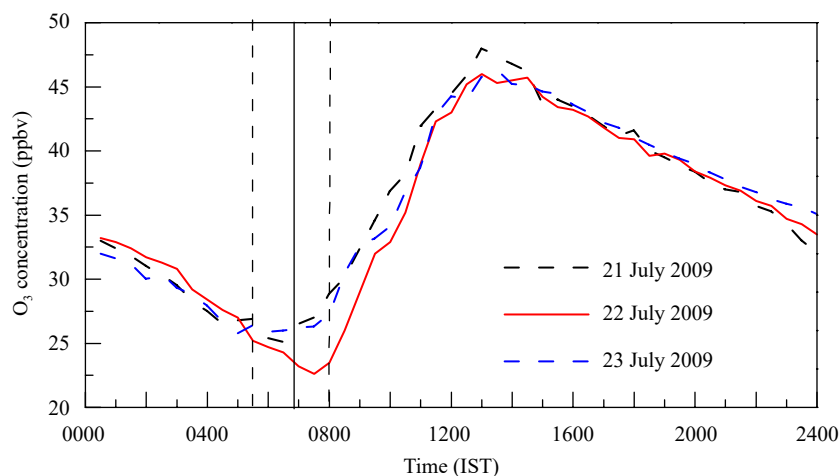


Fig. 3. The diurnal variation of surface O₃ concentration measured on 21, 22, and 23 July 2009 at Kadapa (14.28°N, 78.42°E), India.

ratio of O_3 started to increase again with the increase of solar radiation but remained lower in comparison to the other days. The characteristics of surface O_3 during the solar eclipse mentioned above is related to the photochemical process due to the gradual decrease in the solar radiation affecting the photochemical reactions in the planetary boundary layer (Amiridis et al., 2007). Changes in the mixing ratio of O_3 during the eclipse is expected, because near the surface, it is closely linked with the availability of precursor gases and prevailing meteorological conditions (Nair et al., 2002).

A majority of the experimental investigations of surface O_3 were performed in Europe on 11 August 1999 and 29 March 2006 (e.g., Zerefos et al., 2007; Tzanis et al., 2008; Sharma et al., 2010). Most of the results reported were about the decrease of O_3 concentration during the totality up to 10%–60% and the O_3 minimum delay up to 1 h relatively to the eclipse totality. As a possible mechanism, the disturbance of photochemical activity (photolysis) due to drop of the solar radiation, mainly ultraviolet, is considered. Other important factors are the chemical composition of the atmosphere, cloudiness conditions, vertical and horizontal advection, and diffusion processes between lower and upper layers of the atmosphere.

3.2.2 Impact on NO_2 , NO , and NO_x

In order to visualize fine-scale chemical changes initiated by the solar eclipse, the temporal variations in NO_2 , NO , and NO_x ($NO_2 + NO$) are further examined from 21 to 23 July over Kadapa and shown in Fig. 4. Figure 4a represents the variation of NO_2 while Figs. 4b and 4c represent the variations of NO and NO_x , respectively. The diurnal variation is clearly visible in NO_x and NO_2 while NO does not show any diurnal variation. The diurnal variation in NO_x and NO_2 is opposite to that in O_3 . NO_x and NO_2 mixing ratios are observed to be lower during the day time and higher during the night time. Such diurnal variations are a result of the local emissions, chemical reactions, and boundary layer dynamics at this site. On the contrary to O_3 , the NO_x and NO_2 mixing ratios are observed to increase gradually from the onset of eclipse, reaching their maximum value of 7.5 ppbv during the maximum obscurity phase of eclipse, and decreased thereafter towards the end of eclipse. Changes in the concentration of NO_x and NO_2 observed within the time period of solar eclipse (from the beginning to the end) are suggested to be only associated with the eclipse. However, we could not observe any change in the NO mixing ratio during the eclipse. The increase in NO_x and NO_2 mixing ratios during the eclipse period may be attributed to the reduced photolysis rate of NO_2 because

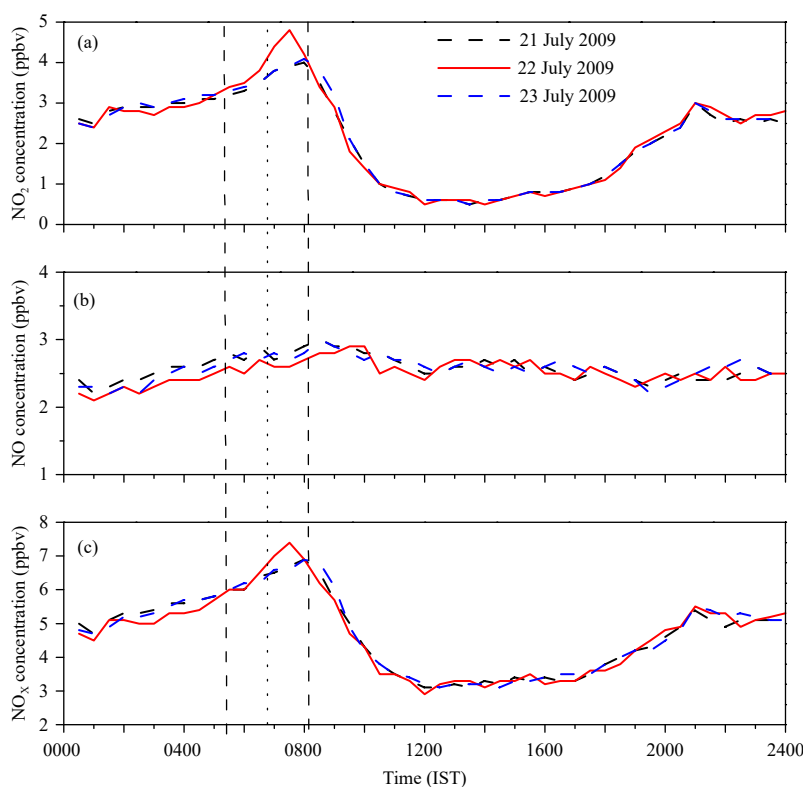
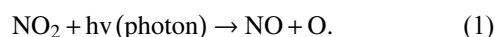


Fig. 4. As in Fig. 3, but for (a) NO_2 , (b) NO , and (c) NO_x .

the observed NO levels did not show any significant change during the entire period of solar eclipse. The photolysis rate of NO₂ can be shown according to the following reaction:



However, an interesting point to be noted is that both the diurnal variation and response to the solar eclipse are opposite in O₃, NO, and NO_x. The average values of O₃ and NO_x mixing ratios are calculated at 30-min time intervals during the solar eclipse period from 0500 to 0800 IST for eclipse as well as control days (as shown in Table 2).

Changes in the solar radiation during the eclipse period can influence the tropospheric O₃ and NO₂ in several ways. The mixing ratio of O₃ is directly affected by the photolysis changes during the eclipse period and indirectly by decrease in the production of O₃ through the photolysis of NO₂. During the solar eclipse period, due to less radiation, the photolysis of NO₂ is reduced, which may result in an increased concentration of NO₂. Current observations support this notion. Details for the O₃ production by the photolysis of NO₂ can be found in Nishanth et al. (2011). The present study on the variation in O₃, NO₂, and NO supports the results of Nishanth et al. (2011) though their study focused on Kannur (11.9°N, 75.4°E) for a different solar eclipse on 15 January 2010. A study by Sharma et al. (2010) for the same solar eclipse (15 January 2010) at Thiruvananthapuram [8.6°N, 76.8°E; 3 m above mean sea level (amsl)] reported a decrease in NO₂ by 69.0%, probably due to the decreased efficiency of the photochemical O₃ formation. But their observational site—Thiruvananthapuram (8.6°N, 76.8°E; 3 m amsl) lies in a relatively polluted environment close to the city (0.5 km away from the national highway—NH 47) and about 6 km from the coast. They have also observed an increase in NO associated with the eclipse induced reduction in O₃, which is attributed to source effects. Another study during the solar eclipse (22 July 2009) at Seoul (37.56°N, 126.98°E) by Kwak et al. (2011) showed contrary results for NO₂ and NO in com-

parison to Sharma et al. (2010). They found enhanced NO₂ and reduced NO because the efficiency of NO₂ photolysis decreased during the eclipse period. The present study supports Kwak et al. (2011) for NO₂ only; however, no perturbations are seen in the NO concentration. The discrepancy in variations of NO₂, NO, and NO_x in current and previous studies (Sharma et al., 2010; Kwak et al., 2011; Nishanth et al., 2011) may be due to different observational sites, suggesting that local factors may dominate during the solar eclipse. As shown in Fig. 2, the wind speed decreases during the solar eclipse, therefore the probability of NO₂ transport will be less. In such cases, the eclipse induced reduction will take place in the photolysis of NO₂. Current results support this mechanism. Thus, it is expected that the chemical production depends upon changes in the local weather and wind induced transport during the eclipse. Further studies are required to quantify the role of geographical factors and effects of NO_x transport by winds during the solar eclipse.

In the next section, the temperature perturbation in the upper tropospheric and ionospheric regions in response to the solar eclipse will be discussed.

3.3 Effects of the solar eclipse on temperature in various atmospheric layers

A preliminary study is conducted on the temperature of cold point tropopause (T-CPT) and the height of CPT (H-CPT) by using radiosonde data. For this purpose, radiosonde measurements at Hyderabad (17.4°N, 78.2°E) from the India Meteorological Department are utilized. The solar eclipse (22 July 2009) had the obscuration of 84% over Hyderabad (Kumar et al., 2013). The spatial distance between Hyderabad and Kadapa is 330 km and the radiosonde launch location is within the vicinity of eclipse. Figure 5 shows the vertical variation of temperature based on the radiosonde measurements for eclipse and control days. In the vicinity of tropopause, slight cooling can be observed during the solar eclipse, especially in the zoomed subplot, which shows a broader peak of the cooling. But these cooling features in T-CPT

Table 2. Average values of O₃ and NO_x concentrations at 30-min time intervals during the period of eclipse (0500–0830 IST 22 July 2009) as well as on control days

Time (IST)	Average O ₃ value on control days	O ₃ value on the eclipse day	Average NO _x value on control days	NO _x value on the eclipse day
0500	28.60	28.30	5.80	5.70
0530	28.40	28.00	6.00	6.00
0600	28.60	25.60	6.10	6.00
0630	28.30	23.80	6.30	6.50
0700	28.90	21.90	6.55	7.00
0730	28.40	20.00	6.65	7.40
0800	28.50	21.50	6.90	6.90
0830	30.30	26.00	6.65	6.20

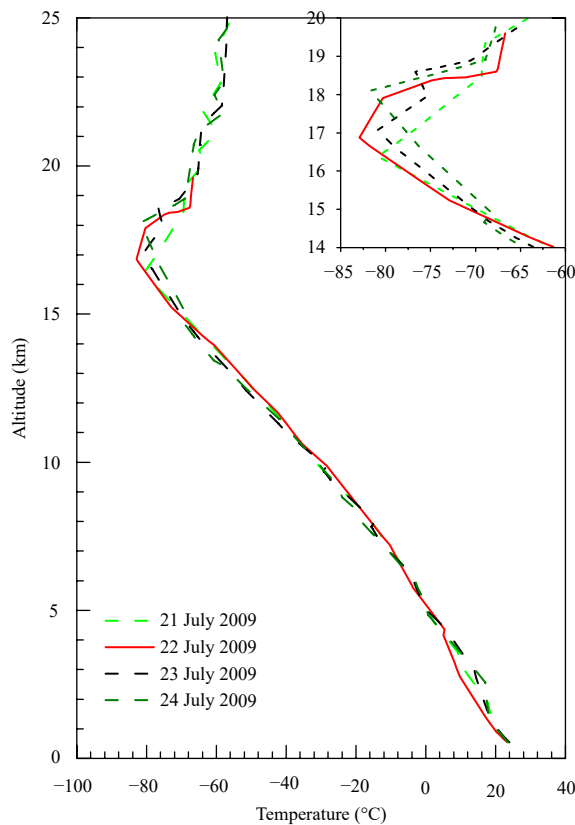


Fig. 5. Daily averaged vertical temperature profiles up to 25 km from 21 to 24 July 2009 based on radiosonde measurements from the India Meteorological Department. The profiles over 14–20 km are zoomed in on the top right corner of the plot.

are not so clear. In addition, Fig. 5 does not reflect any clarity for the variation of T-CPT height. This study does not confirm the cooling pattern in the tropopause region as discussed in previous studies (Bhattacharya et al., 2010; Subrahmanyam et al., 2011). The main reason for this may be the latitudinal location of observational site. Our radiosonde location, i.e., Hyderabad (17.4°N , 78.2°E) does not come directly under the totality of eclipse. Note that in Fig. 5, the vertical variation of temperature is missing for 20 July 2009 because the balloon reached the height of 17.73 km only on this day, which does not cover the entire tropopause region.

To fill up this deficiency and understand the response of tropopause dynamics to eclipse, we further investigate the daily vertical COSMIC temperature data from 20 to 24 July 2009 over the Indian region (65° – 110°E), as shown in Fig. 6. Figure 6a represents the vertical variation of temperature up to 40 km from 20 to 24 July 2009. Figure 6b represents the vertical deviation in temperature perturbation on the solar eclipse day (22 July) in comparison to others days (20, 21, 23, and 24 July). Interestingly, a clear wave fluctuation in the temperature pattern can be seen above 10 km. Note that, on average, the UTLS region showed cooling on the solar eclipse day, opposite to the results of Wang and Liu (2010). By using the COSMIC GPS data between 85°E and 180° , Wang and Liu (2010) observed a warming in the tropopause temperature. However, Wang and Liu (2010) focused on the vast region (85°E – 180°) containing a larger

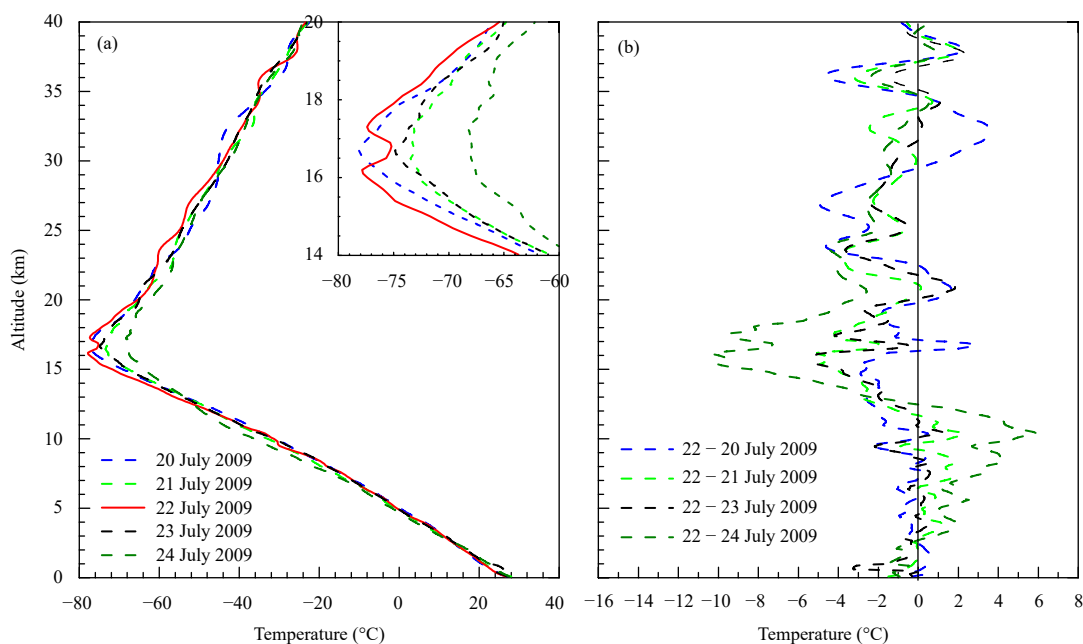


Fig. 6. (a) COSMIC-RO altitude profiles of the daily averaged temperature up to 40 km from 20 to 24 July 2009 over the Indian region (65° – 110°E) and (b) altitude profiles of the temperature perturbation from the eclipse day to control days.

water body part, while our study mainly focused on the land region of India (65° – 110° E). Double tropopause characteristics is quite visible on the eclipse day, which is not noticeable on the other days. The first tropopause level can be seen at 16.2 km with the temperature around -77.84°C , and the second tropopause level at 17.3 km with the temperature around -77.39°C . The variation of temperature at the upper and lower tropopause levels is of the order of 0.39°C only. But it is clearly seen from Fig. 6 that there is a kink between the two height levels (16.2 and 17.3 km), indicting a double tropopause, due to the sharp variation of temperature. The double tropopause characteristic is a new finding from the analysis on the solar eclipse day. The O_3 measurements using balloons and satellites by Randel et al. (2007) showed that profiles with the double tropopause exhibit less O_3 in the stratosphere than those with a single tropopause. The concentration of O_3 in the stratosphere increases by absorption of UV light from the sun. During the solar eclipse, the UV radiation reduces, hence less O_3 production takes place on the eclipse day. Less O_3 formation may be the key component for the formation of double tropopause.

Further variations of T-CPT, H-CPT, and tropical tropopause layer (TTL) are investigated during the control days and eclipse day. Figure 7a represents the daily variation of T-CPT with H-CPT from 20 to 24 July 2009, and Fig. 7b shows the TTL for the same time period. T-CPT appears to drop from the normal height and temperature during the solar eclipse. A cooling of 2°C in T-CPT can be seen with a decrease in height by approximately 400 m. There is no sharp transition from troposphere to stratosphere. The tropopause has a finite width between these two regions, which is generally known as the TTL. CPT is known as an upper boundary of tropopause while the lower boundary is defined by the top of convective outflow/convective tropopause (COT). The region between COT and CPT is known as the TTL width. From Fig. 7b, it is clear that both the upper and lower boundaries of TTL come down simultaneously on

the eclipse day. After the eclipse passed, TTL gets wider. Thus, TTL behaves like a fluid compression and expansion. Note that there is a broader peak observed in the tropopause temperature, hence inducing the wider TTL.

To study the impact of solar eclipse in stratospheric and thermospheric regions, we used the SABER data over the same Indian region (10° – 40°N , 60° – 110°E). The daily vertical variations of SABER temperature profiles from 20 to 24 July are shown in Fig. 8. Figures 8a and 8b represent the temperature variation within 100–130 and 30–110 km, respectively. We could not observe any significant temperature variation in stratospheric and mesospheric regions during the eclipse. But a slight warming in the lower thermospheric region (between 110 and 130 km) can be seen. To refine the finding, we again investigated the daily perturbation in temperature from 20 to 24 July 2009. The average temperature profile for individual day was subtracted from the average of temperature profiles from 20 to 24 except 22 July. To reveal the clear signal of solar eclipse, we did not separate the temperature profile of 22 July 2009. These daily temperature perturbations are shown in Fig. 9. One can notice that temperature in the lower thermosphere shows a variation of about 20 K between 110 and 130 km. Maurya et al. (2014) surmised that GWs were generated in the ionosphere (nearly the 200-km altitude) due to differential heating and in the stratosphere (nearly the 50-km altitude) due to supersonic movement of the moon's shadow. These GWs propagate in both upward and downward directions. The observed drastic heating in the lower thermosphere may be due to breaking of some of these GWs in this region. The drastic heating in the lower thermosphere establishes a strong temperature gradient vertically and horizontally in comparison to the regions that do not have any direct influence of eclipse. Furthermore, this strong temperature gradient may generate new dynamical motions both vertically and horizontally, thus coupling can be initiated in both directions. Therefore, abrupt atmospheric changes may be observed at regions that do not come directly under the path of eclipse. A fur-

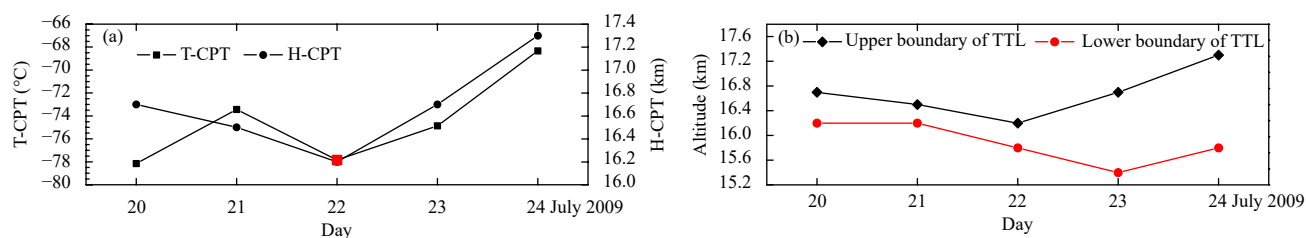


Fig. 7. (a) The temperature of cold point tropopause (T-CPT; $^{\circ}\text{C}$) and the height of CPT (H-CPT; km) and (b) width of tropical tropopause layer (TTL) from 20 to 24 July 2009. (a) and (b) are obtained by using the COSMIC-RO data.

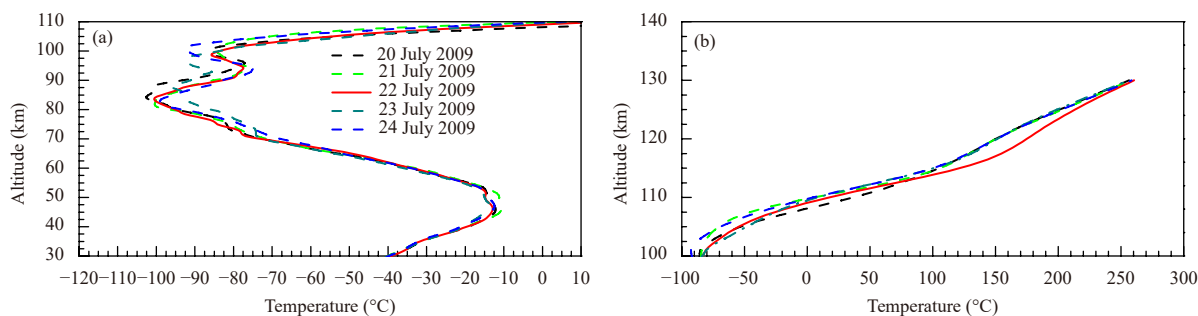


Fig. 8. Temperature profiles over altitudes of (a) 30–110 and (b) 110–140 km from the *TIMED/SABER* measurements.

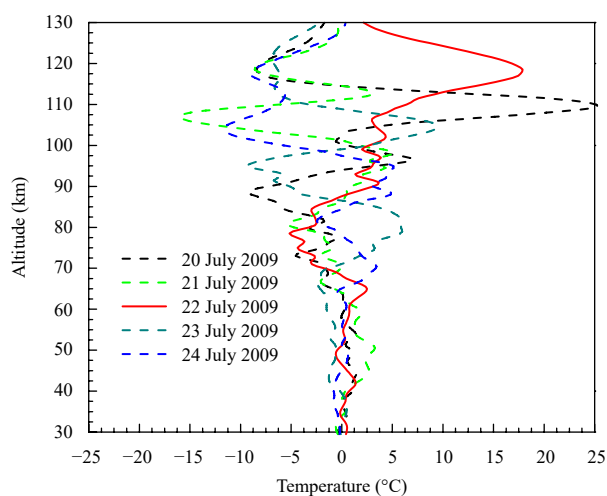


Fig. 9. Altitude profiles of the temperature perturbation of individual day from the average of all temperature profiles during 20–24 July (except 22 July).

ther fine-scale study is required to explore such dynamical coupling by considering several events and using satellite data.

4. Summary

During the total solar eclipse on 22 July 2009, AWS observations over Kadapa showed a strong impact of eclipse on all meteorological and chemical variables. We could not reveal the cooling pattern in the tropopause region based on radiosonde data as discussed in previous studies (Bhattacharya et al., 2010; Subrahmanyam et al., 2011). The location of observations could be the main reason for this discrepancy. The radiosonde location in this study—Hyderabad (17.4°N, 78.2°E) does not come directly under the totality of eclipse whereas the location of previous studies was directly under the totality of eclipse. However, based on the COSMIC satellite data over the Indian region during the eclipse, H-CPT dropped from the normal height by about 400 m and temperature by about 2°C. TTL seems to behave like a fluid compres-

sion and expansion. A double tropopause characteristic is also quite visible on the eclipse day, which has not been explored in previous studies. As discussed above, O_3 concentration changes could be one key parameter for such formation. In addition, the increasing temperature in the lower thermospheric region was found of the order of around 20 K between 110- and 130-km heights. The cooling and heating effects are attributed to the radiative and dynamical responses of the atmosphere to solar eclipse, respectively. Maurya et al. (2014) showed the generation of GWs in the ionosphere and stratosphere and these waves propagate in both upward and downward directions. It may be concluded that significant heating of the lower thermosphere may be a part of the dynamical response to the solar eclipse as breaking of the GWs in this region.

During the solar eclipse period, mixing of the decreased O_3 with increased NO_2 and NO_x were observed. Both the diurnal variation and solar eclipse induced perturbations are contrast in O_3 , NO_2 , and NO_x . We could not observe any impact of solar eclipse on the NO concentration. It is inferred that, during the eclipse period, changes in the local weather and wind induced transport may play a key role in the chemical production variation of species like NO_2 , NO, and NO_x . Future studies are required to quantify the role of geographical factors and effects of NO_x transport by winds during the solar eclipse. These results may have important implications in understanding the response of atmosphere to the radiative, dynamical, and chemical perturbations caused by any celestial or terrestrial disturbance.

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